

A Single Test to Predict Heart Disease Risk

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Introduction of Topic/Problem

The Centers for Disease Control and Prevention's Provisional Mortality Statistics for the year 2024 support a longstanding trend of heart disease as the leading cause of death (Centers for Disease Control and Prevention, 2024). For some people, this may come as a shock, despite the fact that heart disease has been at the number one spot for over a century. According to an American Heart Association survey conducted in November of 2023, "49% of people named heart disease as the leading cause of death; 16% said they didn't know the leading cause, and 18% listed cancer as the top cause of death of people in the U.S" (Lewis, 2024). This means that more than half of U.S. adults don't know that heart disease is the leading cause of death. The data suggests that heart disease will remain number one for 2025, but perhaps predictive analytics can dethrone it in years to come.

Unless you have a pre-existing genetic condition, heart disease doesn't suddenly appear. There are well-known risk factors identifiable in a clinical setting, but they are useless if not identified before the disease's manifestation. You can only receive preventive care if you can afford to visit the doctor; still, "chronic diseases account for most illness, disability, and death in the United States and are the leading drivers of health care costs" (*Fast facts: Health and economic costs of chronic conditions*, 2025). So, not being able to afford preventative care is part of the reason care is so expensive. It's a vicious cycle that needs to be addressed.

This project aims to develop a classification model that scores heart disease risk based on eleven clinical features, enabling prioritized interventions amid cost barriers. When coupled with affordable targeted screening efforts, such as Sanford Health's mobile and center-based screenings, patients can receive personalized risk scores based on historical patient data to inform future health decisions. When risk factors are identified before heart disease manifests, both health care costs and deaths from heart disease and other risk factors can be reduced.

Overview of Data Used

The data was sourced from the Heart Failure Prediction Dataset on Kaggle. The dataset has a 10.0 usability score and was last updated four years ago. It contains 918 observations, 11 distinct features, and a binary classification target variable indicating whether the patient has heart disease. The features included are age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, resting electrocardiogram results, maximum heart rate achieved, exercise-induced angina, old peak, and the slope of the peak exercise ST segment. These features are relevant to the problem because they assess both heart anomalies and risk factors for heart disease. The dataset is a combination of five distinct datasets collected from Cleveland, OH, Hungary, Switzerland, Long Beach, VA, and the Stalog Heart Data Set.

Methods of Analysis

Preliminary analysis suggested that the dataset is well-suited to the problem being solved. The dataset didn't contain any obvious missing values or duplicates, so all 918 rows could be used for modeling and testing. Definitions provided alongside the Kaggle dataset for each variable enhanced understanding of its features. The dataset required minimal data preparation. After minor imputation on a single variable, skew correction for two variables, scaling of numeric variables, and one-hot encoding of categorical features, the dataset was ready for model selection and fitting.

RandomizedSearchCV was used to select the best classification model and hyperparameters, as it's less computationally intensive than GridSearchCV while still producing meaningful results. The four models tested were XGBoost Classifier, Support Vector Classification (SVC), Random Forest Classifier, and Logistic Regression. These models were chosen for their ability to produce a probability score for the positive class, "Has heart disease," which can then be further classified into low, medium, and high-risk

levels, each used for targeted clinical advice. The models were selected using the scoring metric, 'neg_brier_score', which "measures the mean squared difference between the predicted probability and the actual outcome" (*Brier_score_loss*, n.d.). This metric was chosen to optimize probability calibration, ensuring that each patient receives appropriate guidance based on their risk level.

Results & Findings Explained

One of the many advantages of using RandomizedSearchCV is that it stores the models that have already been created for further evaluation. Each model was evaluated on four different scoring metrics.

Table I

RandomizedSearchCV Model Evaluation Metrics

Model	Brier_Score	Log_Loss	ROC_AUC	F1_Score
XGBoost	0.0863	0.2855	0.9571	0.8900
SVC	0.0972	0.3162	0.9482	0.8812
Logistic Regression	0.0990	0.3196	0.9470	0.8744
Random Forest	0.0961	0.3170	0.9448	0.8856

Table I shows that the XGBoost model outperformed the other three models across all metrics. It succeeded in both calibration (lowest Brier score: 0.0863) and discrimination (highest ROC AUC: 0.9571). It also delivered the most trustworthy probability outputs (lowest Log Loss: 0.2855), and the best binary classification balance (F1 Score: 0.8900). Despite XGBoost's superiority, the scores were strong across all models.

With an efficient pipeline that funnels users' raw data, all of the models evaluated can yield interpretable, actionable results. Though in the spirit of Occam's Razor, the simplest model with a comparable result is usually the best choice. Feature selection was not originally part of the plan, since

only 11 features were included in the dataset, but the feature importance values for the XGBoost model made it clear that ST_Slope_Up carried most of the predictive value. In fact, its importance value was over 9 times that of the next-highest value, ExerciseAngina_Y.

Table II

Top Heart Disease Predictors

Top Heart Disease Predictors:		
	feature	importance
14	ST_Slope_Up	0.566793
12	ExerciseAngina_Y	0.062507
5	Sex_M	0.043213

Because of the importance finding, RandomizedSearchCV was again implemented using only the top three features, and its evaluation metrics were compared with those of the original models with all 11 features. While losing both calibration and discriminative power, the scores from the reduced features remained comparable to those of the 11-feature models. This result suggests that simplifying the model via feature selection does not greatly impact its predictive power.

Table III

RandomizedSearchCV Model Evaluation Metrics (With Top Three Feature Models)

Model	Brier_Score	Log_Loss	ROC_AUC	F1_Score
XGBoost	0.0863	0.2855	0.9571	0.8900
SVC	0.0972	0.3162	0.9482	0.8812
Logistic Regression	0.0990	0.3196	0.9470	0.8744
Random Forest	0.0961	0.3170	0.9448	0.8856
XGBoost Top 3	0.1225	0.3966	0.9040	0.8529
SVC Top 3	0.1348	0.4384	0.9040	0.8529
Random Forest Top 3	0.1224	0.3968	0.9040	0.8529
Logistic Regression Top 3	0.1225	0.3935	0.8983	0.8529

Additionally, when comparing the results for the three feature models, XGBoost no longer dominates all categories. Random Forest Top 3 has the best calibration (lowest Brier_Score: 0.1224), while Logistic Regression most successfully penalizes incorrectly predicted probabilities (lowest Log_Loss 0.3935). Logistic Regression also has the ROC_AUC score of 0.8983 compared to every other model's score of 0.9040. All models yielded the same harmonic mean of precision and recall (F1: 0.8529).

In practice, a three-feature model can significantly reduce the time and cost required to run multiple tests and perform data preparation, while still maintaining strong predictive power. Patients wouldn't need to have blood drawn for testing, which would save time and money on supplies and labor. Not to mention, for patients who are afraid of needles, a blood test could be part of the reason they don't want to complete a screening in the first place. If this barrier is removed, patients with a fear of needles may be more likely to participate. By focusing on only three features, the imputation and skew adjustment steps can be omitted during data preparation, simplifying the model's pipeline.

The following table shows the clinical risk distribution and examples of tiered risk scores for the test data using the three-feature Logistic Regression model.

Table IV

Clinical Risk Distribution & Score Examples

Clinical Risk Distribution on Testing Data:

```
High      109
Low       24
Medium    51
Name: count, dtype: int64
```

Score Examples for First 10 Predictions:

```
0.928 → High
0.740 → High
0.051 → Low
0.928 → High
0.928 → High
0.191 → Medium
0.517 → High
0.928 → High
0.191 → Medium
0.191 → Medium
```

Conclusion

The three-feature Logistic Regression model essentially matches the performance of the three-feature XGBoost Ensemble model. Logistic Regression models are excellent choices for clinical deployment because of their interpretability. Not only can the model provide the necessary clinical risk scoring for proper patient counseling post-screening, but clinicians also have the added benefit of communicating odds ratios to their patients, specific to each feature. For example, when a clinician reports an odds ratio of 0.08 for the ST_Slope_Up feature, they can say that an upsloping ST is associated with 92% lower odds of heart disease. For these reasons, this analysis recommends using the three-feature Logistic Regression model for predicting heart disease risk scores.

It's important to note that the most predictive feature in this model, the ST_Slope treadmill test, is often the longest and most expensive test to administer. In a standard test, the patient is placed on a treadmill (or bike), and about 12 electrocardiogram (ECG) leads are attached to the patient's skin to record measurements as the patient progresses through increasingly difficult levels of exercise. From these

measurements, the ST_Slope is calculated and categorized into either up, down, or flat. The flat and down categorizations correlate most strongly with heart disease.

Trying to identify the majority of heart disease risk from just one or two leads could further streamline screening time and costs. One 2024 deep learning study's analysis aimed to "identify fine-grained ECG features from deep learning, which are then used to derive reduced-ECG leads for a variety of tasks" (Lu et al., 2024). Screening agents could begin by asking whether the patient has pain while exercising or is sedentary (in which case they wouldn't be able to honestly answer the question), and whether the patient identifies as male, and make ST_Slope screening suggestions based on those two indicators. They could also offer a screening to anyone who wants one, based on the level of funding.

By streamlining both mobile and center-based screening efforts to focus on only one test, the ST_Slope feature, and by conducting further model evaluations using previously published studies to reduce the number of leads required for the test, clinicians can use the three-feature Logistic Regression model to deliver tiered and interpretable results in real time. Additionally, deploying the model in these screening centers would provide an excellent opportunity to collect additional training data for underrepresented classes, thereby strengthening the model's performance on all demographics over time.

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